

Smart Gym Management System

Web & Mobile Application Features + Hardware/Software Requirements

Introduction

This document lists the Web Application and Mobile Application features proposed for the Smart Gym Management System at Asian Fitness Center, plus the hardware and software needed to deliver it. It serves as an initial scoping reference to confirm coverage before development begins.

The system runs on a Laravel backend, with a React web application for the Owner, Staff, and Coach, and a React Native mobile application for Members.

Web Application Features

The web application serves the Owner, Coach roles, each with access scoped to their responsibilities.

Owner

- Dashboard view of daily gym status (members in, income, unpaid members, stock levels)
- Member management (add, edit, view profiles)
- Payment management (Daily and Monthly payment tracking, Stock buying and selling)
- Staff/Coach management
- Offers & discounts management (birthday, renewal, win-back, couple package)
- Special message broadcasts (birthday, offers, announcements)
- Fridge & food stock management
- Gym products sales (Sale → Bill → Payment → Stock update)
- Bills and machine maintenance records
- Gym reports and analytics
- Remote access — check gym status from anywhere

Coach

- Member management (registration, profile updates)
- Payment recording and tracking
- Gym products sales processing
- Daily task / job management checklist (cleaning, equipment checks, stock checks)
- View and update assigned members' Exercise Plans and Diet Plans
- Track member progress (weight, calories, exercise progress)
- View daily fitness-related tasks
- Shift-based coach schedule visibility

Cross-cutting (Admin/system-wide)

- Gym entry control tied to payment status (entry logged/flagged, not device-blocked)
- Attendance tracking (today, weekly, monthly, per-member)
- Member review and feedback tracking

Mobile Application Features

The mobile app is Member-facing, giving each member full self-service visibility into their own gym journey.

- Personal profile
- Payment history
- Attendance history
- Diet Plan (today's meals, meal times, recommended foods, coach notes)
- Exercise Plan (daily workouts, sets, mark exercises complete)
- Daily calorie tracking
- Weight tracking
- Progress view (weight, calories, exercise, monthly trend)
- Reviews (submit feedback on service, coach, or plans)
- Offers & discounts
- Messages (birthday greetings, announcements, special offers)
- Coach information (assigned coach, shift/availability)
- Zumba class schedule, attendance, and class information

Hardware Requirements

Item	Purpose	Qty
ZKTeco SenseFace 3A (facial recognition entry terminal)	Physical gym entry control and identification at the entrance	1
Portable printer	Printing bills for product/gym sales at the counter	Customer-supplied
Staff-facing device (existing PC/tablet at front desk)	Running the web application for staff and owner use	Customer-supplied

Software Requirements

Component	Details
Backend	Laravel (API, authentication, role-based access, business logic)
Web application	React (Owner, Staff, Coach interfaces)
Mobile application	React Native (Member app, Android & iOS)
Hosting	Cloud server hosting for the backend, database, and dashboard/reports
Software development	Development, configuration, deployment, testing, and staff training
Maintenance	Software updates and on-call technical support included in the monthly hosting fee